

13

Quantum computing

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¹Quantum computing requires precise control of the motion of the ions, i.e. their external degrees of freedom, as well as their internal state $|F, M_F\rangle$.

Quantum computing will be a revolutionary new form of computation in the twenty-first century, able to solve problems inaccessible to classical computers. However, building a quantum computer is very difficult, and so far only simple logic gates have been demonstrated in experiments on ions in a linear Paul trap. The ideas of quantum computing have also been tested in experiments using nuclear magnetic resonance (NMR).

In the Paul trap, the ions sit in an ultra-high vacuum (pressure $\sim 10^{-11}$ mbar) so that collisions rarely happen, and the ions are well isolated from the environment. We have seen that these conditions enable extremely high-resolution spectroscopy of single ions because of the very small perturbations to the energy levels. Quantum computing requires more than one ion in the trap, and all of these ions must be cooled to the lowest vibrational level to give a well-defined initial quantum state for the system.¹ This presents a much greater experimental challenge than the laser cooling of a single ion to reduce Doppler broadening, but it has been achieved in some experiments. The previous chapter describes the physics of the linear Paul trap but for the purposes of this chapter we simply assume that the trap produces a harmonic potential with strong confinement in the radial direction, so that the ions lie in a line along its z -axis, as shown in Fig. 13.1; their mutual electrostatic repulsion keeps the ions far enough apart for them to be seen separately.

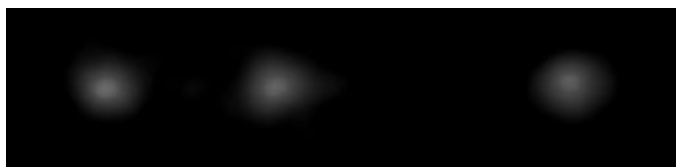


Fig. 13.1 A string of *four* ions in a linear Paul trap. Coulomb repulsion keeps the ions apart and the gap corresponds to an ion in the dark state. In this experiment two laser beams simultaneously excite strong and weak transitions to give quantum jumps, as described in Section 12.6, so that each ion flashes on and off randomly. This snapshot of the system at a particular time could be taken to represent the binary number 1101. A quantum logic gate requires much more sophisticated techniques in which laser pulses determine precisely the initial state of each individual ion in the chain, as described in this chapter. Courtesy of Professors A. M. Steane and D. N. Stacey, D. M. Lucas and co-workers, Physics department, University of Oxford.

13.1 Qubits and their properties

A classical computer uses bits with two values 0 or 1 to represent binary numbers, but a quantum computer stores information as quantum bits or qubits (pronounced Q-bits). Each qubit has two states, labelled $|0\rangle$ and $|1\rangle$ in the Dirac ket notation for quantum states. Most theoretical discussions of quantum computing consider the qubit as a spin-1/2 object, so the two states correspond to spin down $|m_s = -1/2\rangle$ and spin up $|m_s = +1/2\rangle$. However, for a trapped ion the two states usually correspond to two hyperfine levels of the ground configuration, as illustrated in Fig. 13.2. In the following discussion, $|1\rangle$ represents the ion in the upper hyperfine level and $|0\rangle$ is used for the lower hyperfine level, but all of the arguments apply equally well to spin-1/2 particles since the principles of quantum computing clearly do not depend on the things used as qubits. Ions, and other physical qubits, give a compact way of storing information, e.g. $|1101\rangle$ represents the binary number 1101 in Fig. 13.1, but the quantum features of this new way of encoding information only become apparent when we consider the properties of more than one qubit in Section 13.1.1. Even though a single qubit generally exists in a superposition of the two states, a qubit does not carry more classical information than a classical bit, as shown by the following argument. The superposition of the two states

$$\psi_{\text{qubit}} = a|0\rangle + b|1\rangle \quad (13.1)$$

obeys the normalisation condition $|a|^2 + |b|^2 = 1$. We write this superposition in the general form

$$\psi_{\text{qubit}} = \left\{ \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \right\} e^{i\phi'}. \quad (13.2)$$

The overall phase factor has little significance and the possible states correspond to vectors of unit length with direction specified by two angles θ and ϕ . These are the position vectors of points lying on the surface of a sphere, as in Fig. 13.3. The state $|0\rangle$ lies at the north pole of this *Bloch sphere* and $|1\rangle$ at the south pole; all other position vectors are superpositions of these two basis states. Since these position vectors correspond to a simple classical object such as a pointer in three-dimensional space, it follows that the information encoded by each qubit can be modelled in a classical way.

The analogy between a qubit and a three-dimensional pointer seems to imply that a qubit stores more information than a bit with two possible values 0 or 1, e.g. like one of the hands of an analogue clock that gives us information about the time by its orientation in two-dimensional space. Generally, however, this is not true because we cannot determine the orientation of a quantum object as precisely as the hands on a clock. Measurements can only distinguish quantum states, with a high probability, if the states are very different from each other, occupying well-separated positions on opposite sides of the Bloch sphere. A measurement on a

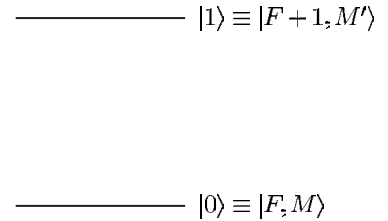
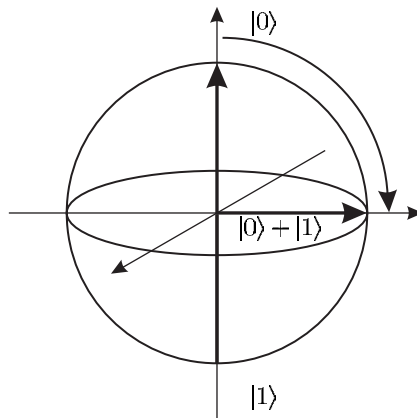


Fig. 13.2 The two hyperfine levels in the ground state of an ion. Generally, experiments use an ion with total angular momentum $J = 1/2$ in the lowest electronic configuration, so that $F = I \pm 1/2$. The qubits $|0\rangle$ and $|1\rangle$ correspond to two particular Zeeman states in the two levels, e.g. $|F, M\rangle$ and $|F+1, M'\rangle$, respectively.

Fig. 13.3 The state $|0\rangle$ lies at the north pole of this *Bloch sphere* and $|1\rangle$ at the south pole; all other position vectors are superpositions of these two basis states. The Bloch sphere lies in the Hilbert space spanned by the two eigenvectors $|0\rangle$ and $|1\rangle$. The Hadamard transformation defined in eqn 13.3 takes $|0\rangle \mapsto |0\rangle + |1\rangle$ (cf. Fig. 7.2).



spin-1/2 particle determines whether its orientation is up or down along a given axis. After that measurement the particle will be in one of those states since the act of measurement puts the system into an eigenstate of the corresponding operator. In the same way, a qubit will give either 0 or 1, and the read-out process destroys the superposition.

The Bloch sphere is very useful for describing how individual qubits transform under unitary operations. For example, the *Hadamard* transformation that occurs frequently in quantum computation (see the exercises at the end of this chapter) has the operator²

$$\begin{aligned}\hat{U}_H |0\rangle &\mapsto |0\rangle + |1\rangle, \\ \hat{U}_H |1\rangle &\mapsto |0\rangle - |1\rangle.\end{aligned}\tag{13.3}$$

This is equivalent to the matrix

$$\hat{U}_H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The effect of this unitary transformation of the state is illustrated in Fig. 13.3—it corresponds to a rotation in the Hilbert space containing the state vectors. This transformation changes $|0\rangle$, at the north pole, into the superposition given in eqn 13.3 that lies on the equator of the sphere.

13.1.1 Entanglement

We have already encountered some aspects of the non-intuitive behaviour of multi-particle quantum systems in the detailed description of the two electrons in the helium atom (Chapter 3), where the antisymmetric spin state $[|\downarrow\uparrow\rangle - |\uparrow\downarrow\rangle]/\sqrt{2}$ corresponds to the wavefunction

$$\Psi = |01\rangle - |10\rangle\tag{13.4}$$

in the notation used in this chapter (without normalisation). This does not factorise into a product of single-particle wavefunctions:

$$\Psi \neq \psi_1 \psi_2,\tag{13.5}$$

²In this chapter wavefunctions are written without normalisation, which is the common convention in quantum computation.

where

$$\psi_1\psi_2 = [a|0\rangle + b|1\rangle]_1 [c|0\rangle + d|1\rangle]_2. \quad (13.6)$$

Here c and d are additional arbitrary constants, and the futility of attempting to determine these constants quickly becomes obvious if you try it. Generally, we do not bother with the subscript used to denote the particle, so $|0\rangle_1|1\rangle_2 \equiv |01\rangle$ and $|1\rangle_1|1\rangle_2 \equiv |11\rangle$, etc. Multiple-particle systems that have wavefunctions such as eqn 13.4 that cannot be written as a product of single-particle wavefunctions are said to be **entangled**. This entanglement in systems with two, or more, particles leads to quantum properties of a completely different nature to those of a system of classical objects—this difference is a crucial factor in quantum computing. Quantum computation uses qubits that are *distinguishable*, e.g. ions at well-localised positions along the axis of a linear Paul trap. We can label the two ions as Qubit 1 and Qubit 2 and know which one is which at any time. Even if they are identical, the ions remain distinguishable because they stay localised at certain positions in the trap. For a system of distinguishable quantum particles, any combination of the single-particle states is allowed in the wavefunction of the whole system:

$$\Psi = A|00\rangle + B|01\rangle + C|10\rangle + D|11\rangle. \quad (13.7)$$

The complex amplitudes A , B , C and D have arbitrary values. It is convenient to write down wavefunctions without normalisation, e.g.

$$\Psi = |00\rangle + |01\rangle + |10\rangle + |11\rangle, \quad (13.8)$$

$$\Psi = |00\rangle + 2|01\rangle + 3|11\rangle, \quad (13.9)$$

$$\Psi = |01\rangle + 5|10\rangle. \quad (13.10)$$

Two of these three wavefunctions possess entanglement (see Exercise 13.1). We encounter examples with three qubits later (eqn 13.12).

In the discussion so far, entanglement appears as a mathematical property of multiple-particle wavefunctions, but what does it mean physically? It is always dangerous to ask such questions in quantum mechanics, but the following discussion shows how entanglement relates to correlations between the particles (qubits), thus emphasising that entanglement is a property of the system *as a whole* and not the individual particles. As a specific example consider two trapped ions. To measure their state, laser light excites a transition from state $|1\rangle$ (the upper hyperfine level) to a higher electronic level to give a strong fluorescence signal, so $|1\rangle$ is a ‘bright state’, while an ion in $|0\rangle$ remains dark.³ Wavefunctions such as those in eqns 13.4 and 13.10 that contain only the terms $|10\rangle$ and $|01\rangle$ always give one bright ion and the other dark, i.e. an anticorrelation where a measurement always finds the ions in different states. To be more precise, this corresponds to the following procedure. First, prepare two ions so that the system has a certain initial wavefunction Ψ_{in} , then make a measurement of the state of the ions by observing their fluorescence. Then reset the system to Ψ_{in} before another measurement. The record of the state of the ions for a sequence

³This is similar to the detection of quantum jumps in Section 12.6, but typically quantum computing experiments use a separate laser beam for each ion to detect them independently.

of such measurements looks like: 10, 10, 01, 10, 10, ... Each ion gives a random sequence of 0s and 1s but always has the opposite state to the other ion.

This example does not illustrate the full subtlety of entanglement because we would get the same result if we prepared the two ions either in $|01\rangle$ or $|10\rangle$ randomly at the beginning. Such an apparatus produces correlated pairs of ions in a purely classical way that mimics the quantum situation. John Bell proved that we can make measurements that distinguish the ‘classical correlation’ from an entangled state. The above description shows that this cannot be done simply by making measurements along the axes defined by the basis states $|0\rangle$ and $|1\rangle$, but it turns out that quantum entanglement and ‘classically-correlated’ particles give different results for measurements along other sets of axes. This was a very profound new insight into the nature of quantum mechanics that has stimulated much important theoretical and experimental work.⁴

⁴Bell considered the well-known EPR paradox and the system of two ions described above has the same properties as the two spin-1/2 particles usually used in quantum mechanics texts (Rae 1992).

‘Entanglement implies correlation but correlation does not imply entanglement’. In the following we shall concentrate mainly on the first half of this statement, i.e. two-particle systems encode quantum information as a *joint* property of the qubits and carry more information than can be stored on the component parts separately. The quantum information in an entangled state is very delicate and is easily destroyed by perturbations of the relative phase and amplitude of the qubits, e.g. in present-day ion traps it is difficult to maintain coherence between more than a few qubits; this decoherence is caused by random perturbations that affect each qubit in a different way.

The wavefunctions of the two electrons in helium are entangled but they do *not* give qubits useful for quantum computing. Nevertheless, since this is a book about atomic physics it is worthwhile to look back at helium. The antisymmetric state of the two spins has already been used as an example of entanglement. The symmetric spin wavefunction $[|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle]/\sqrt{2}$ also has entanglement, but the two other symmetric wavefunctions factorise: $|\uparrow\uparrow\rangle \equiv |\uparrow\rangle_1|\uparrow\rangle_2$, and similarly for $|\downarrow\downarrow\rangle$. The two electrons are in these eigenstates of S because of the exchange symmetry. When the two electrons do not have the same quantum numbers, n and l , the spatial wavefunctions are symmetric and antisymmetric combinations of the single-electron wavefunctions and are entangled. These eigenstates of the residual electrostatic interaction also satisfy the requirement of exchange symmetry for identical particles. (Note that the energy levels and spatial wavefunctions would be the same even if the particles were not identical—see Exercise 3.4.) The exchange integrals in helium can be regarded as a manifestation of the entanglement of the spatial wavefunction of the two electrons that leads to a correlation in their positions, or an anticorrelation, making it more (or less) probable that the electrons will be found close together. From the quantum perspective, the energy difference for two different entangled wavefunctions does not seem strange because we do not expect them to have the same properties, even if they are made up of the same single-electron states.

13.2 A quantum logic gate

Quantum computing uses nothing more than standard quantum mechanics but it combines operations on the qubits and quantum measurements in sophisticated ways to give amazingly powerful new methods for computation. Here we consider just one example of the transformations of the qubits, or quantum logic gates, that form the elementary building blocks (primitives) of a quantum computer. The controlled-NOT gate (CNOT) transforms two qubits, as shown in Table 13.1.

The CNOT gate changes the value of the second qubit *if and only if* the first qubit has the value 1. The first qubit controls the effect of the gate on the second qubit. The truth Table 13.1 looks the same as that for ordinary logic, but a quantum logic gate corresponds to an operation that preserves the superposition of the input states. The quantum mechanical operator of the CNOT gate $\hat{U}_{\text{CNOT}}$ acts on the wavefunction of the two qubits to give

$$\begin{aligned} \hat{U}_{\text{CNOT}} \{A|00\rangle + B|01\rangle + C|10\rangle + D|11\rangle\} \\ \rightarrow A|00\rangle + B|01\rangle + C|11\rangle + D|10\rangle. \end{aligned} \quad (13.11)$$

Alternatively, the effect of this operation can be written as $|10\rangle \leftrightarrow |11\rangle$, with the other states unchanged. The complex numbers A , B , C and D represent the phase and relative amplitude of states within the superposition.

13.2.1 Making a CNOT gate

The CNOT gate is elementary but it turns out not to be the simplest gate to make in an ion trap quantum processor—it requires a sequence of several operations, as described in Steane (1997). So, despite the fact that quantum computing has been introduced into this book as an application of ion trapping, this section explains how to make a quantum logic gate for two spin-1/2 particles. This two-qubit system is not just a convenient theoretical case, but corresponds to actual quantum computing experiments carried out with the nuclear magnetic resonance (NMR) technique. Figure 13.4 shows the energy levels of the two interacting spins.

We do not need to go into all the details of how this energy-level structure arises to understand how to make a CNOT gate, but it is important

Table 13.1 Truth table of the CNOT quantum logic gate.

$ 00\rangle \rightarrow 00\rangle$
$ 01\rangle \rightarrow 01\rangle$
$ 10\rangle \rightarrow 11\rangle$
$ 11\rangle \rightarrow 10\rangle$

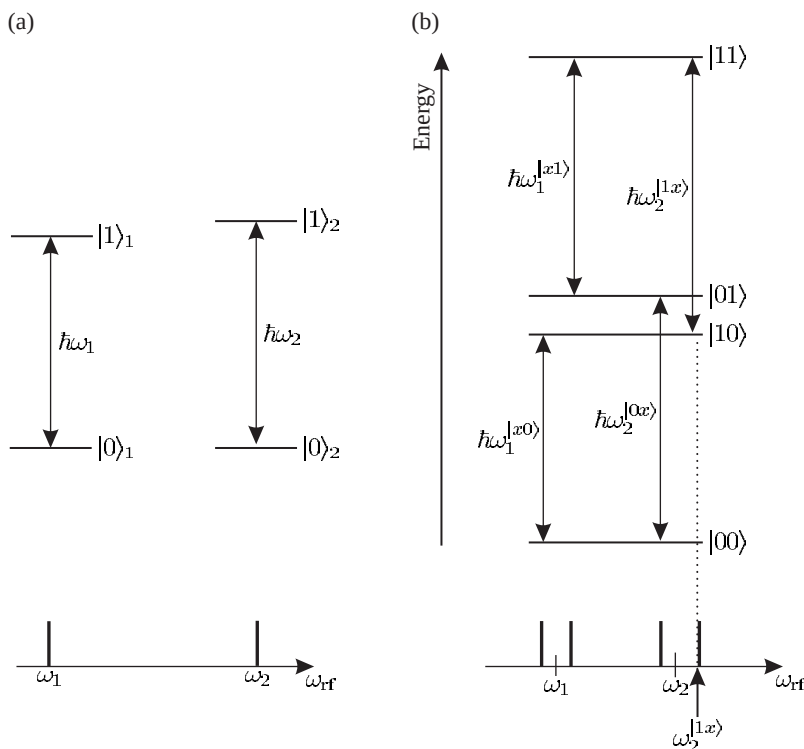


Fig. 13.4 (a) The energy levels of two spin-1/2 particles which do not interact with each other. The frequencies that drive the transition between the levels of Qubit 1 and Qubit 2 are ω_1 and ω_2 , respectively. In NMR these levels correspond to the up and down orientations of two protons in a strong magnetic field, i.e. the states $|m_I = \pm 1/2\rangle$ for each proton. The difference in the resonance frequencies $\omega_1 \neq \omega_2$ arises from interaction with their surroundings (other nearby atoms in the molecule). (b) The energy levels of two interacting particles, or qubits. The interaction between the qubits makes the frequency required to change the orientation of Qubit 2 depend on the state of the other qubit (and similarly for Qubit 1). To denote this, the two new resonance frequencies close to ω_2 are labelled with the superscripts $|0x\rangle$ and $|1x\rangle$. (Note this rather cumbersome notation is not generally used but it is useful in this introductory example.) The absorption spectrum is drawn below the corresponding transitions. A π -pulse of radio-frequency radiation at angular frequency ω_2^{1x} switches Qubit 2 ($|0\rangle_2 \leftrightarrow |1\rangle_2$) if and only if Qubit 1 is in $|1\rangle_1$. This gives the CNOT gate of Table 13.1. For details of the NMR technique see Atkins (1994). In NMR the interaction of the nuclear magnetic moment with a strong external magnetic field $g_I \mu_B \mathbf{I} \cdot \mathbf{B}$ dominates. For a field of 10 T the protons at the centre of the hydrogen atoms in the sample have resonance frequencies of 400 MHz. This frequency corresponds to approximately $\omega_1/2\pi \simeq \omega_2/2\pi$ in the figure. The *chemical shift* causes ω_1 and ω_2 to differ by only a few parts per million, but this is resolved by standard NMR equipment. Atkins (1994) describes how the fine structure in NMR spectra, shown in (b), arises from spin–spin coupling between the two nuclear spins.

⁵The amplitudes and phases of the states within the superposition must not change spontaneously, i.e. the decoherence must be very small over the time-scale of the experiment.

to know that spontaneous emission is negligible.⁵ To make our discussion less abstract, it is also useful to know that in NMR experiments the energy differences between the levels shown in Fig. 13.4 arise from the orientation of the magnetic moment of the nuclei, proportional to their spin, in a strong magnetic field and that radio-frequency radiation drives transitions between the states. There is a splitting of $\hbar\omega_1$ between the up and down states of the first qubit ($|0\rangle$ and $|1\rangle$) and $\hbar\omega_2$ for the second qubit. Therefore a pulse of radio-frequency radiation at angular frequency ω_1 changes the orientation of the first spin (Qubit 1); for example, a π -pulse (as defined in Section 7.3.1) swaps the states of this

qubit $|0\rangle \leftrightarrow |1\rangle$, i.e. $a|0\rangle + b|1\rangle \mapsto a|1\rangle + b|0\rangle$. Similarly, pulses at angular frequency ω_2 manipulate the state of the other spin (Qubit 2). These changes in the state of each qubit independently are not sufficient for quantum computing. The qubits must interact with each other so that one qubit ‘controls’ the other qubit and influences its behaviour. A small interaction between the spin causes the shift of the energy levels indicated in going from Fig. 13.4(a) to (b)—the energy required to flip Qubit 2 now depends on the state of Qubit 1 (and vice versa). The energy levels in Fig. 13.4(b) give four separate transition frequencies between the four states of the two qubits; a pulse of radio-frequency radiation that drives one of these transitions can achieve the CNOT operation—it changes the state of Qubit 2 only if Qubit 1 has state $|1\rangle$.⁶

Although it seems simple for the spin-1/2 system, the implementation of a quantum logic gate represents a major step towards building a quantum computer. Rigorous proofs based on mathematical properties of unitary operators in quantum mechanics show that any unitary operation can be constructed from a few basic operators—it is sufficient to have just one gate which gives control of one qubit by another, such as the CNOT, in addition to the ability to manipulate the individual qubits in an arbitrary way.⁷ These operators form a so-called universal set which generates all other unitary transformations of the qubits.

⁶The coupling between trapped ions arises from the mutual electrostatic repulsion of the ions, but this interaction between qubits does *not* give a level structure like that in Fig. 13.4.

⁷To rotate the nuclear spins to an arbitrary point on the Bloch sphere the NMR experiments use radio-frequency pulses. Experiments in ion traps use Raman pulses (Section 9.8).

13.3 Parallelism in quantum computing

A classical computer acts on binary numbers stored in the input register, or registers, to output another number also represented as bits with values 0 and 1. A quantum computer acts on the whole superposition of all the input information in the qubits of its input register, e.g. a string of ions in a linear Paul trap. This quantum register can be prepared in a superposition of all possible inputs at the same time, so that the quantum computing procedure transforms the entire register into a superposition of all possible outputs. For example, with $N = 3$ qubits a general initial state containing all possible inputs has the wavefunction

$$\begin{aligned} \Psi = & A|000\rangle + B|001\rangle + C|010\rangle + D|011\rangle \\ & + E|100\rangle + F|101\rangle + G|110\rangle + H|111\rangle. \end{aligned} \quad (13.12)$$

Quantum computing corresponds to carrying out a transformation, represented by the quantum mechanical operator $\hat{U}$, to give the wavefunction $\Psi' = \hat{U}\Psi$ that is a superposition of the outputs corresponding to each input

$$\Psi = A\hat{U}|000\rangle + B\hat{U}|001\rangle + C\hat{U}|010\rangle + \dots \quad (13.13)$$

Useful quantum algorithms combine different operations, such as $\hat{U}_{\text{CNOT}}$, to give an overall transformation $\hat{U} = \hat{U}_m \cdots \hat{U}_2 \hat{U}_1$, and these operations have the same advantage of parallelism as for the individual operations. It appears that, instead of laboriously calculating the output for each of the binary numbers 000 to 111, corresponding to 0

to 7 in decimals, the quantum computation does all eight calculations simultaneously. Herein lies the great power of quantum computing.

The number of combinations of the input states increases exponentially with the number of qubits; for a register of 100 qubits this means a transformation of $2^{100} \equiv 10^{30}$ inputs in parallel—an astronomical number. But how can the correct answer be picked out from the enormous number of possible outputs? All the possibilities encoded in the multiple-particle wavefunction Ψ' *cannot* be turned into a list of the outputs for a given input, as given by a classical computer. Indeed, a simple quantum measurement just gives one of the outputs chosen randomly from all the possibilities. However, quantum mechanics allows measurements that give some information about all of the outputs, if we renounce the possibility of getting information about any particular output—just as the complementarity principle stops us knowing both the position and momentum of a particle precisely. As a trivial example that gives the flavour of this idea, but is in fact not strictly correct, consider the measurement of the state of the last qubit in the output register—if all the possible outputs are even numbers then this ion will be in $|0\rangle$. So we can establish something about all the possibilities in a single quantum computation, but this does not allow any more information to be extracted, i.e. if they are not all even we cannot find out which ones are odd (nor which input states gave these outputs). Similar ideas apply for more complicated quantum algorithms, e.g. finding the prime factors of numbers. (The real nature of quantum computation can only properly be appreciated by working through a quantum example, so the grossly simplified case given here must not be pushed too far.)

The factorisation of large numbers is often quoted as a ‘killer application’ of quantum computing, i.e. something that they can do which cannot be done on existing computers (within a realistic time). It is obviously easy to multiply two prime numbers together, e.g. $37 \times 61 = 2257$; but it is much more difficult to go in the opposite direction, e.g. to find the prime factors of 1271 (try it for yourself). For the larger numbers a classical computer can easily do the multiplication, but the factorisation of a number of order 10^{100} would take an exceedingly long time even with the fastest supercomputers and the most efficient classical algorithm to search for the prime factors. It has been proved rigorously that quantum algorithms exist to attack this type of problem—the algorithms act like a filter only letting through the sought-after combination of qubits from the input superposition state. The practical impossibility of factorising large numbers on present-day computers forms the foundation of the best methods of encryption.⁸ A quantum computer has the ability to try many combinations in each computation and so quickly find the key. The possibility of cracking the best currently-used codes has prompted such hard-headed bodies as the government security agencies to investigate quantum computing and fund research on ion traps. One wonders what they think of the intrigue and mystery of quantum mechanics.

⁸The need for safe methods of encrypting information has always been crucial for the military, and nowadays it is important for confidential electronic transfer of data in business and to protect credit card numbers.

13.4 Summary of quantum computers

Qubits are quantum objects that store information and can exist in arbitrary superpositions $a|0\rangle + b|1\rangle$. A quantum computer is a set of qubits on which the following operations can be carried out.

- (1) Each qubit can be prepared in a given state, so the quantum registers of the computer have a well-defined initial state.
- (2) In quantum computation the quantum logic gates (unitary transformations) act on a chosen subset of the qubits. During these processes the system is in an entangled state where the information is encoded in the state of the entire quantum register. This cannot be reduced to a description of terms of the individual qubits, like a list of 0s and 1s for a classical computer. Some of these operations are control operations in which the change in the state of a qubit depends on the state of other qubits.
- (3) The final state of the qubits is read out by making a quantum measurement.

For quantum information processing with a string of trapped ions, the three stages of initialisation, quantum logic and read out correspond to the following operations:

- (1) Preparation of the initial state—all of the ions must be cooled to the ground vibrational state and be in the same internal state. Any state $|F, M_F\rangle$ of a given hyperfine level will do and the choice is based on practical considerations.⁹
- (2) Raman transitions change both the internal and the vibrational states of the ions (qubits) to implement the operation of quantum gates.
- (3) Laser beams resonant with a strong transition determine which hyperfine level each ion is in at the end of the process (Section 12.6). The ions lie at least $10\text{ }\mu\text{m}$ apart, so that each one is seen individually (as in Fig. 13.1).

⁹In the NMR experiments the preparation of the initial state, or the resetting of the device after a computation, are not straightforward since those systems do not have a dissipative process equivalent to laser cooling.

Quantum computing needs only a few basic types of quantum logic gate. A control gate between a single pair of qubits in a multiple-particle system can be combined with swap operations to effectively extend the operation to all pairs of qubits. These manipulations combined with arbitrary rotations of individual qubits give a universal set of operators from which all other unitary operators can be constructed.

13.5 Decoherence and quantum error correction

The power of quantum computing to solve important problems has been proved mathematically, but so far experiments have only used

¹⁰At that time NMR experiments used up to seven qubits.

a few qubits to demonstrate the principle of some elementary operations. These experiments all use state-of-the-art techniques, but for large systems the external perturbations blur the entangled states so that they cannot be distinguished from each other, e.g. for ions in the Paul trap the electrical noise on the radio-frequency electrodes causes random changes of the phase of the states within a superposition. In the year 2000 the best ion trap experiment was limited to four qubits.¹⁰ Decoherence quickly causes a system of many qubits to become so muddled up that it is impossible to pick out the required output. In contrast, in a classical computer errors which change 0 into 1, or the other way round, occur extremely rarely, and present-day computers have procedures for correcting errors so that they have a negligible effect.

Decoherence was thought to be an incurable disease in real quantum systems that would prevent quantum information processing from ever being carried out with enough qubits to make it really useful. However, a new cunning way of encoding information has been invented that cures the symptoms of decoherence. So-called quantum error correction (QEC) exploits the subtle features of the theory of quantum mechanics to get rid of the small amounts of the unwanted states that gradually get mixed into the states containing the quantum information by perturbations on the system. Broadly speaking, making a certain type of quantum measurement causes the wavefunction of the system of qubits to ‘collapse’, in a way that destroys any additional phase or amplitude introduced by decoherence. This measurement must be made with respect to a very carefully chosen basis of eigenstates to preserve the entanglement of the wavefunction. Clearly, it is not adequate to make a simple measurement that causes quantum superpositions to collapse into just one of the states within the superposition, since this would completely destroy the coherence between the states that we want to preserve.¹¹

In the context of ion traps, the recent advances in quantum computing and QEC can be regarded as the latest steps in the evolution of spectroscopy and laser cooling, as illustrated by the following rough history.

¹¹QEC manages to make measurements on eigenstates that are in some sense orthogonal to those in the original superposition. These special quantum measurements cause the small admixture of other states introduced into the wavefunction by the noise to collapse to zero. The QEC measurements are performed on extra qubits entangled with the qubits in the quantum register; measurements made on these *ancillary* qubits do not reduce the size of the superposition state that stores quantum information (in the register), i.e. the number of qubits in the superposition is preserved in QEC. The technical details of this process are explained in Steane (1998). Some elementary aspects of QEC can be appreciated by analogy with the quantum Zeno effect described in Exercise 13.5.

- (1) The first spectroscopists observed light from discharge lamps, e.g. the Balmer lines in atomic hydrogen. They used spectrographs and étalons and the resolution was limited by Doppler broadening, together with collisions and other broadening mechanisms in the discharge.
- (2) Atomic beams allowed experimenters to change the population between the various energy levels of the atoms, using radio-frequency or microwave radiation to manipulate hyperfine and Zeeman levels in the ground state. (Optical pumping was applied to the ground states of certain atoms.) Atoms were deflected slightly in the Stern–Gerlach experiment but without a significant change in speed. The laser extended these techniques to the higher levels using optical transitions, so that, in principle, atomic physics experiments can manipulate the internal states of the atoms and put atoms into any

desired energy level.

- (3) Ion traps confined the ions for long periods with buffer gas cooling.
- (4) The advent of laser cooling of atoms meant that spectroscopists were no longer passive observers, but could magnetically trap neutral atoms and control the atom's motion. Laser cooling of ions can put them into the ground state of the trap to give a completely defined external state.
- (5) Quantum computing techniques allow the manipulation of both the external and internal state of the trapped ion qubits with laser light, to put a system of many ions into a particular quantum state.
- (6) Quantum error correction can be regarded as a refined form of laser cooling. QEC puts the ions back into a coherent superposition of the desired states of the system (actually a particular sub-space of Hilbert space), rather than the lowest energy state as in conventional cooling. It was a real surprise that decoherence in quantum mechanics can be overcome in this way and actively stabilise quantum information.

13.6 Conclusion

At heart, quantum mechanics remains mysterious. Certain aspects of quantum behaviour seem strange to our intuition based on the classical world that we experience directly, e.g. the well-known examples of 'which way the photon goes' in a Young's double-slit experiment and Schrödinger's cat paradox.¹² These examples have provoked much discussion and thought over the years about the so-called quantum measurement problem. Nowadays, physicists do not regard the peculiarities of quantum systems as a problem at all, but rather as an opportunity. A proper appreciation of the profoundly different properties of multiple-particle systems and the nature of entanglement has shown how to use their unique behaviour in quantum computing. Just as a car mechanic uses her practical knowledge to get an engine working, without worrying about the details of internal combustion, so a quantum mechanic (or quantum state engineer) can design a quantum computer using the known rules of quantum mechanics without worrying too much about philosophical implications. Physicists, however, strive towards a better understanding of the quantum world and consideration of the profound and subtle ideas in quantum information theory sheds new light on aspects of quantum mechanics. In physics research there often exists a symbiotic relationship between theory and experimental work, with each stimulating the other; a prime example is quantum error correction which was not thought about until people started to face up to the fact that none of the systems used in existing experiments has any possibility of working reliably with a useful number of qubits (without error correction)—of course, the problem of decoherence has always been recognised but experimental results focused attention on this issue.

¹²See quotation from Heisenberg in the background reading section of the Preface.

The theoretical principles of quantum computing are well understood but there are many practical difficulties to overcome before it becomes a reality. All potential systems must balance the need to have interactions between the qubits to give coherent control and the minimisation of the interactions with the external environment that perturb the system. Trapped ions have decoherence times much longer than the time needed to execute quantum logic gates and this makes them one of the most promising possibilities. In quantum computing many new and interesting multiple-particle systems have been analysed and, even if they cannot be realised yet, thinking about them sharpens our understanding of the quantum world, just as the EPR paradox did for many years before it could be tested experimentally.

Further Reading

The *Contemporary Physics* article by Cummins and Jones (2000) gives an introduction to the main ideas and their implementation by NMR techniques. The books by Nielsen and Chuang (2000) and Stolze and Suter (2004) give a very comprehensive treatment. The article on the ion-trap quantum information processor by Steane (1997) is also useful background for Chapter 12. Quantum computing is a fast-moving field with new possibilities emerging all the time. The latest information can be found on the World-Wide Web.

Exercises

(13.1) Entanglement

- (a) Show that the two-qubit state in eqn 13.8 is *not* entangled because it can be written as a simple product of states of the individual particles in the basis $|0'\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ and $|1'\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.
- (b) Write the maximally-entangled state $|00\rangle + |11\rangle$ in the new basis.
- (c) Is $|00\rangle + |01\rangle - |10\rangle + |11\rangle$ an entangled state? [*Hint.* Try to write it in the form of eqn 13.6 and find the coefficients.]
- (d) Show that the two states given in eqns 13.9 and 13.10 are both entangled.

- (e) Discuss whether the three-qubit state $\Psi = |000\rangle + |111\rangle$ possesses entanglement.

(13.2) Quantum logic gates

This question goes through a particular example of the statement that any operation can be constructed from a combination of a control gate and arbitrary rotations of the individual qubits. For trapped ions the most straightforward logic gate to build is a controlled ‘rotation’ of Qubit 2 when Qubit 1 is $|1\rangle$, i.e.

$$\begin{aligned}\hat{U}_{\text{CROT}} \{A|00\rangle + B|01\rangle + C|10\rangle + D|11\rangle\} \\ = A|00\rangle + B|01\rangle + C|10\rangle - D|11\rangle.\end{aligned}$$

- (a) Write $\hat{U}_{\text{CROT}}$ as a 4×4 matrix and show that it is unitary.
- (b) Equation 13.3 defines the Hadamard transformation. Write this transformation for Qubit 2, $\hat{U}_H(2)$ as a 4×4 matrix with the same basis states as in (a), and show that it is unitary.
- (c) Find the combination of $\hat{U}_{\text{CROT}}$, $\hat{U}_H(2)$ and $\hat{U}_H^\dagger(2)$ that gives the CNOT gate (Table 13.1).
- (13.3) *Elementary operations with more than two qubits*
The CNOT gate where the state of Qubit 1 controls whether Qubit 2 switches (Table 13.1) corresponds to the operator $\hat{U}_{\text{CNOT}}(1, 2)$. Note that $\hat{U}_{\text{CNOT}}(1, 2) \neq \hat{U}_{\text{CNOT}}(2, 1)$ and that these gates do not affect other qubits.
- (a) A system of three qubits has a quantum bus that enables operations that swap the states of any pair of qubits: $\hat{U}_{\text{SWAP}}(1, 2)$ for $1 \leftrightarrow 2$, $\hat{U}_{\text{SWAP}}(2, 3)$ for $2 \leftrightarrow 3$ and $\hat{U}_{\text{SWAP}}(1, 3)$ for $1 \leftrightarrow 3$. What combination of $\hat{U}_{\text{CNOT}}(1, 2)$ and SWAP operators gives $\hat{U}_{\text{CNOT}}(1, 3)$, a CNOT gate where Qubit 1 controls Qubit 3?
- (b) The operator $\hat{U}_H(i)$ gives the transformation $|0\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$ for the i th qubit. Show that $\hat{U}_H(3)\hat{U}_H(2)\hat{U}_H(1)$ acting on $|000\rangle$ gives the three-qubit state with all coefficients equal, namely $A = B = C = D = E = F = G = H$ in eqn 13.12.
- (c) The three operations in part (b) prepare eight input states, i.e. they put a three-qubit register into a superposition of the eight states. How many inputs are prepared by thirty such operations on thirty qubits. Is the initial state of the quantum register prepared in this way entangled?
- (13.4) *Grover's search algorithm*
Imagine trying to find a given number in the telephone directory when you have written down the number but not the name. There is no clever classical way of speeding up this tedious task but the massive parallelism of a quantum algorithm does make a big difference to this sort of problem. In

this question we consider finding a particular value of a two bit number x from the list of four possibilities, but the principles can be extended to larger numbers. To implement the search the quantum computer uses an operation in which the sign of the input changes for the specified value—if we assume the answer is $x = |11\rangle$ then the operation causes $|11\rangle \mapsto -|11\rangle$ but leaves the other states unchanged. Instead of a simple list, the assignment of x could be made on the basis of a solution to an algebraic problem, e.g. the operator changes the sign of the input x if some function of x has a particular value such as $f(x) = 0$. In order that the computer can implement the sign change efficiently it is only necessary that the function $f(x)$ can be evaluated efficiently for some general x . This is a much easier task than finding the value of x for which $f(x) = 0$.

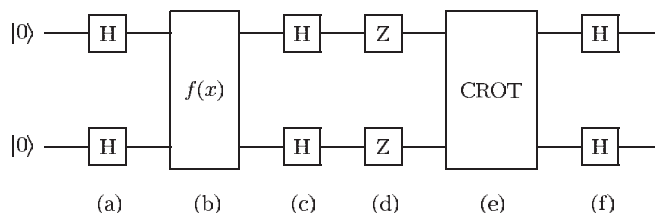
The figure shows the sequence of operations corresponding to Grover's algorithm, starting with two qubits in $|00\rangle$. The operators are defined in the following. The letters (a) to (f) indicate the part of this question corresponding to each stage. (Ignore normalisation factors throughout this question.)

- (a) Show that the Hadamard transformation (eqn 13.3) applied to each of the qubits of $|00\rangle$ results in the superposition state

$$\Psi_{\text{in}} = |00\rangle + |01\rangle + |10\rangle + |11\rangle.$$

This is the initial state of the two qubits needed for the algorithm.

- (b) The box labelled $f(x)$ represents the operation for the function, as described above. In this question this corresponds to a CROT gate (defined in Exercise 13.2). Write down $\hat{U}_{\text{CROT}}\Psi_{\text{in}}$.
- (c) The next step in the algorithm is another Hadamard transformation of each qubit. We have already worked out the effect on $|00\rangle$ in part (a). Work out the effect on $|01\rangle$, $|10\rangle$ and $|11\rangle$. Show that the superposition is $|00\rangle + |01\rangle + |10\rangle - |11\rangle$.



- (d) The operator $\hat{U}_Z$ corresponds to $|0\rangle \mapsto |0\rangle$ and $|1\rangle \mapsto -|1\rangle$, for each qubit. Write down the resulting state of the system of two qubits.
- (e) Show that after another $\hat{U}_{\text{CROT}}$ the state is $|00\rangle - |01\rangle - |10\rangle + |11\rangle$.
- (f) Show that a final Hadamard transformation yields the required answer.
- (g) Repeat the algorithm for another function with solution $x = 10$, to show that it picks out the required value, i.e. $f(x)$ corresponds to $|10\rangle \mapsto -|10\rangle$.

Comment. This algorithm has a similar ‘one-way’ nature to the factorisation of numbers mentioned in the text—it is very time consuming to find the prime factors, but once possible solutions have been found by the quantum computation it is easy to verify whether they are indeed factors (or satisfy the equation) using a classical computer. Some more complex quantum algorithms do not always give the ‘correct’ answer but they are still useful in ‘winnowing the wheat from the chaff’; the algorithms pick out all the required answers (grains of wheat) along with a few unwanted numbers. This is not a problem since invalid numbers are thrown away after checking—the set of numbers to be checked is much smaller than the initial set of all possibilities, so the procedure is efficient.

(13.5) *The quantum Zeno effect*

The ancient Greek philosopher Zeno proposed various arguments against motion, e.g. *Achilles and the tortoise* in which the logic leads to the paradoxical conclusion that the man cannot overtake the tortoise. In the quantum Zeno effect the evolution of a wavefunction is slowed down by repeated quantum measurements on the system, and this has been described by the colloquial phrase ‘a watched pot never boils’.

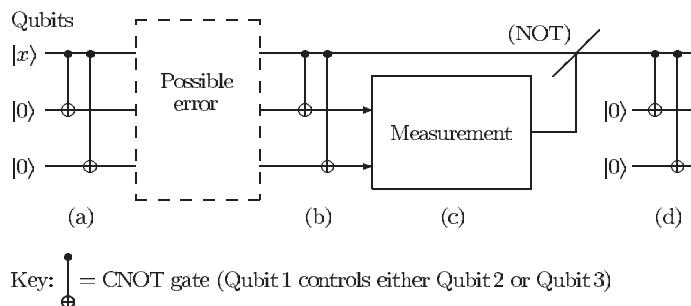
This question is based on the usual treatment of Rabi oscillations rewritten in terms of the states of a qubit that starts in $|0\rangle$ at time $t = 0$. Transitions between the states are induced by a perturbation which is characterised by the Rabi frequency Ω :

$$|\psi\rangle = \cos\left(\frac{\Omega t}{2}\right) |0\rangle - i \sin\left(\frac{\Omega t}{2}\right) |1\rangle.$$

- (a) The state of the qubit is measured after a short time $\tau \ll 1/\Omega$. Show that the probability that the qubit ends up in $|1\rangle$ is $(\Omega\tau/2)^2 \ll 1$. What is the probability that the qubit is in $|0\rangle$ after the measurement?
- (b) The state of the qubit is measured at time $\tau/2$. What are now the probabilities of $|0\rangle$ and $|1\rangle$?
- (c) After the measurement at time $\tau/2$ the qubit will be in either $|0\rangle$ or $|1\rangle$. The qubit then evolves for time $\tau/2$ and its state is measured again at time τ . Calculate the probability of the outcome $|1\rangle$.
- (d) For a sequence of n measurements with a time interval of τ/n between them, the probability of having made a transition from $|0\rangle$ to $|1\rangle$ at time τ is $1/n$ times that in part (a). Verify this result for $n = 3$ or, if you can, justify it for the general case.
- (e) Discuss the application of these results to the measurement of the frequency of a narrow transition by the quantum jump technique, as described in Section 12.6. The transition rate on the weak transition decreases as the measurement periods of excitation on the strong transition become more frequent. What effect does this have on the measured line width of the narrow transition?

(13.6) *Quantum error correction*

Classical computers encode each number with more than the minimum number of bits necessary; the additional ‘check’ bits allow the system to detect whether any of the bits has undergone a random change (caused by noise). In these error correction codes the binary code for each number differs from the string of 0s and 1s for any other possible number by at least two bits, so a random change of a single bit leads to an invalid binary code that the computer rejects. In practice, error correction codes have the binary strings representing valid entries ‘further apart’ to protect against errors in several bits. Quantum error correction (QEC) uses extra qubits to make the computational states ‘further apart’ (in Hilbert space), so the system is more robust, i.e. it is harder for the computational states to become mixed.



The above figure shows a particular QEC scheme. On the left-hand side, Qubit 1 starts in $|x\rangle$ and Qubits 2 and 3 both start in $|0\rangle$. The letters (a) to (d) indicate the part of this question corresponding to each stage.

- (a) To represent the qubit $|x\rangle = a|0\rangle + b|1\rangle$ this quantum error correction code uses $\Psi_{\text{in}} = a|000\rangle + b|111\rangle$. Show that two CNOT gates acting on Qubits 2 and 3, with Qubit 1 as the control qubit in both cases, as indicated in the figure, encode $|x\rangle$ in this way.
- (b) Suppose that a perturbation of Qubit 1 causes a bit-change error so that the state of the three qubits becomes $\alpha(a|000\rangle + b|111\rangle) + \beta(a|100\rangle + b|011\rangle)$. Here β is the amplitude of the unwanted state mixed into the original state (and normalisation determines α). Show that after another two CNOT gates we get $(a|0\rangle + b|1\rangle)\alpha|00\rangle + (a|1\rangle + b|0\rangle)\beta|11\rangle$.

- (c) The measurement of the states of Qubits 2 and 3 has two possible results. One possibility is to find $|11\rangle$, which means that Qubit 1 has changed and in this case the error is corrected by applying a (conditional) NOT operation to Qubit 1, i.e. $|0\rangle \leftrightarrow |1\rangle$ if and only if the measurement of the other qubits gives $|11\rangle$. What is the other possible state of Qubits 2 and 3 resulting from the measurement? Verify that Qubit 1 ends up in the original state $|x\rangle$ after this stage.
- (d) Qubits 2 and 3 are both reset to $|0\rangle$. Two further CNOT gates recreate Ψ_{in} from $|x\rangle$, just as in stage (a).

This scheme corrects ‘bit flip’ errors in any of the qubits. Show this by writing out what happens in stages (b), (c) and (d) for the state

$$\alpha(a|000\rangle + b|111\rangle) + \beta(a|100\rangle + b|011\rangle) + \gamma(a|010\rangle + b|101\rangle) + \delta(a|001\rangle + b|110\rangle).$$

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.

A

Appendix A: Perturbation theory

A.1 Mathematics of perturbation theory	298
A.2 Interaction of classical oscillators of similar frequencies	299

Although degenerate perturbation theory is quite simple (cf. the series solution of differential equations used to find the eigenenergies of the hydrogen atom), it is often regarded as a ‘difficult’ and subtle topic in introductory quantum mechanics courses. Degenerate perturbation theory arises often in atomic physics, e.g. in the treatment of helium (eqn 3.14). Equations 6.34 and 7.89 in the treatment of the Zeeman effect on hyperfine structure and the a.c. Stark effect, respectively, also have a similar mathematical form, as can be appreciated by studying the general case in this appendix. Another aim of this appendix is to underline the point made in Chapter 3 that degenerate perturbation theory is not a mysterious quantum mechanical phenomenon (associated with exchange symmetry) and that a similar behaviour occurs when two classical systems interact with each other.

A.1 Mathematics of perturbation theory

The Hamiltonian for a system of two levels of energies E_1 and E_2 (where $E_2 > E_1$) with a perturbation given by H' , as in eqn 3.10, can be written in matrix form as

$$H_0 + H' = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} + \begin{pmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{pmatrix}. \quad (\text{A.1})$$

The matrix elements of the perturbation are $H'_{12} = \langle \psi_1 | H' | \psi_2 \rangle$, and similarly for the others. The expectation values $\langle \psi_1 | H' | \psi_1 \rangle$ and $\langle \psi_2 | H' | \psi_2 \rangle$ are the usual first-order perturbations. It is convenient to write the energies in terms of a mean energy J and an energy interval 2ϵ that takes into account the energy shift caused by the diagonal terms of the perturbation matrix:

$$\begin{aligned} E_1 + H'_{11} &= J - \epsilon, \\ E_2 + H'_{22} &= J + \epsilon. \end{aligned}$$

For simplicity, we assume that the off-diagonal terms are real, e.g. as for the case of exchange integrals in helium and the other examples in this book:¹

$$H'_{12} = H'_{21} = K.$$

¹This is normally the case when levels are bound states. We shall find that the eigenenergies depend on K^2 , which would generalise to $|K|^2$.

When $\epsilon = 0$ this leads to a matrix eigenvalue equation similar to that used in the treatment of helium (eqn 3.14), but the formalism given here enables us to treat both degenerate and non-degenerate cases as different limits of the same equations. This two-level system is described by the matrix equation

$$\begin{pmatrix} J + \epsilon & K \\ K & J - \epsilon \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = E \begin{pmatrix} a \\ b \end{pmatrix}. \quad (\text{A.2})$$

The eigenenergies E are found from the determinantal equation (as in eqn 7.91 or eqn 3.17):

$$E = J \pm \sqrt{\epsilon^2 + K^2}. \quad (\text{A.3})$$

This exact solution is valid for all values of K , not just small perturbations. However, it is instructive to look at the approximate values for weak and strong interactions.

(a) *Degenerate perturbation theory, $K \gg 2\epsilon$*

If $K \gg 2\epsilon$ then the levels are effectively degenerate, i.e. their energy separation is small on the scale of the perturbation. For this strong perturbation the approximate eigenvalues are

$$E = J \pm K, \quad (\text{A.4})$$

as in helium (Section 3.2). The two eigenvalues have a splitting of $2K$ and a mean energy of J . The eigenfunctions are admixtures of the original states with equal amplitudes.

(b) *Perturbation theory, $K \ll 2\epsilon$*

When the perturbation is weak the approximate eigenvalues obtained by expanding eqn A.3 are

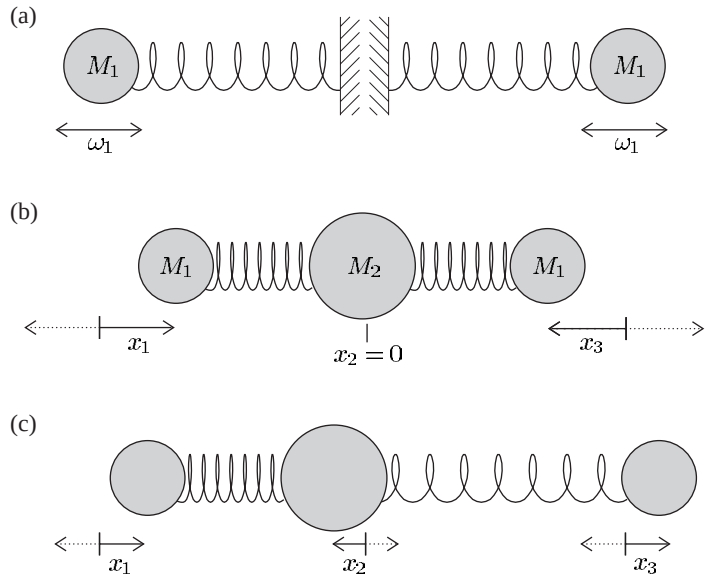
$$E = J \pm \left(\epsilon + \frac{K^2}{2\epsilon} \right). \quad (\text{A.5})$$

This is a second-order perturbation, proportional to K^2 , as in eqns 6.36 and 7.92.

A.2 Interaction of classical oscillators of similar frequencies

In this section we examine the behaviour of two classical oscillators of similar frequencies that interact with each other, e.g. the system of two masses joined by three springs shown in Fig. 3.3, or alternatively the system of three masses joined by two springs shown in Fig. A.1. The mathematics is very similar in both cases but we shall study the latter because it corresponds to a real system, namely a molecule of carbon dioxide (with M_1 representing oxygen atoms and M_2 being a carbon

Fig. A.1 An illustration of degenerate perturbation theory in classical mechanics, closely related to that shown in Fig. 3.3. (a) The ‘unperturbed’ system corresponds to two harmonic oscillators, each of which has a mass M_1 attached to one end of a spring and with the other end fixed. These independent oscillators both have the same resonant frequency ω_1 . (b) A system of three masses joined by two springs corresponds to two coupled harmonic oscillators. When $M_2 \gg M_1$ the coupling is weak. The displacements x_1 , x_2 and x_3 are measured from the rest position of each mass (and are taken as being positive to the right). One eigenmode of the system corresponds to a symmetric stretch in which the central mass does not move; therefore this motion has frequency ω_1 (independent of the value of M_2). (c) The asymmetric stretching mode has a frequency higher than ω_1 . (This system gives a simple model of molecules such as carbon dioxide; a low-frequency bending mode also arises in such molecules.)



atom). The equations of motion for this ‘ball-and-spring’ molecular model are

$$M_1 \ddot{x}_1 = \kappa(x_2 - x_1), \quad (\text{A.6})$$

$$M_2 \ddot{x}_2 = -\kappa(x_2 - x_1) + \kappa(x_3 - x_2), \quad (\text{A.7})$$

$$M_1 \ddot{x}_3 = -\kappa(x_3 - x_2), \quad (\text{A.8})$$

where κ is the spring constant (of the bond between the carbon and oxygen atoms). The x -coordinates are the displacement of the masses from their equilibrium positions. Adding all three equations gives zero on the right-hand side since all the internal forces are equal and opposite and there is no acceleration of the centre of mass of the system. This constraint reduces the number of degrees of freedom to two. It is convenient to use the variables $u = x_2 - x_1$ and $v = x_3 - x_2$. Substituting $\kappa/M_1 = \omega_1^2$ and $\kappa/M_2 = \omega_2^2$ leads to the matrix equation

$$\begin{pmatrix} \ddot{u} \\ \ddot{v} \end{pmatrix} = \begin{pmatrix} -(\omega_1^2 + \omega_2^2) & \omega_2^2 \\ \omega_2^2 & -(\omega_1^2 + \omega_2^2) \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (\text{A.9})$$

A suitable trial solution is

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} e^{-i\omega t}. \quad (\text{A.10})$$

This leads to an equation with the same form as eqn A.2 with $\epsilon = 0$ (cf. eqn 3.14).

The determinantal equation yields

$$(\omega_1^2 - \omega^2) (\omega_1^2 + 2\omega_2^2 - \omega^2) = 0, \quad (\text{A.11})$$

giving the two eigenfrequencies $\omega = \omega_1$ and $\omega' = \sqrt{\omega_1^2 + 2\omega_2^2}$. For frequency ω_1 the eigenvector is $b = a$, corresponding to a symmetric stretch (Fig. A.1(b)); this is the same as the initial frequency since M_2 does not move in this normal mode. The other normal mode of higher frequency corresponds to an asymmetric stretch with $b = -a$ (see Fig. A.1(c)). In these motions the centre of mass does not accelerate, as can be verified by summing eqns A.6, A.7 and A.8.

A treatment of two coupled oscillators of different frequencies is given in Lyons (1998) in the chapter on normal modes; this is the classical analogue of the non-degenerate case that was included in the general treatment in the previous section. The books by Atkins (1983, 1994) give a comprehensive description of molecules.

B

Appendix B: The calculation of electrostatic energies

¹The notation indicates that the electrons are labelled 1 and 2, meaning that their locations are called $\mathbf{r}_1$ and $\mathbf{r}_2$. At the same time, the electron charges are (probably) differently distributed in space (because the electrons' wavefunctions have different quantum numbers), so ρ_1 and ρ_2 are different functions of their arguments.

²We have called the integral J in anticipation of working out a direct integral, but an exchange integral is equally well catered for if we use appropriate formulae for $\rho_1(\mathbf{r}_1)$ and $\rho_2(\mathbf{r}_2)$.

³The integrals are written with the integrand at the end to make clear the range of integration for each variable.

This appendix describes a method of dealing with the integrals that arise in the calculation of the electrostatic interaction of two electrons, as in the helium atom and other two-electron systems. Consider two electrons¹ whose charge densities are $\rho_1(\mathbf{r}_1)$ and $\rho_2(\mathbf{r}_2)$. Their energy of electrostatic repulsion is (cf. eqn 3.15)

$$J = \iint \rho_1(\mathbf{r}_1) \frac{e^2}{4\pi\epsilon_0 r_{12}} \rho_2(\mathbf{r}_2) d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad (\text{B.1})$$

In this expression we shall leave open the precise form of the two charge densities, so the theorem we are to set up will apply equally to direct and to exchange integrals.² Also, we shall permit the charge density $\rho_1(\mathbf{r}_1) = \rho(r_1, \theta_1, \phi_1)$ to depend upon the angles θ_1 and ϕ_1 , as well as upon the radius r_1 , and similarly for $\rho_2(\mathbf{r}_2)$; neither charge density is assumed to be spherically symmetric.

In terms of the six spherical coordinates the integral takes the form

$$J = \int_0^\pi d\theta_1 \sin \theta_1 \int_0^{2\pi} d\phi_1 \int_0^\pi d\theta_2 \sin \theta_2 \int_0^{2\pi} d\phi_2 \\ \times \int_0^\infty dr_1 r_1^2 \rho_1(r_1, \theta_1, \phi_1) \int_0^\infty dr_2 r_2^2 \rho_2(r_2, \theta_2, \phi_2) \frac{e^2}{4\pi\epsilon_0 r_{12}}.$$

In rearranging this expression, we process just the two radial integrals. Within them, we divide the range for r_2 into a part from 0 to r_1 and a part from r_1 to infinity. The radial integrals become³

$$\begin{aligned}
 & \int_0^\infty dr_1 r_1^2 \rho_1(\mathbf{r}_1) \int_0^{r_1} dr_2 r_2^2 \rho_2(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 r_{12}} \\
 & + \int_0^\infty dr_1 r_1^2 \rho_1(\mathbf{r}_1) \int_{r_1}^\infty dr_2 r_2^2 \rho_2(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 r_{12}} \\
 & = \int_0^\infty dr_1 r_1^2 \rho_1(\mathbf{r}_1) \int_0^{r_1} dr_2 r_2^2 \rho_2(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 r_{12}} \\
 & \quad + \int_0^\infty dr_2 r_2^2 \rho_2(\mathbf{r}_2) \int_0^{r_2} dr_1 r_1^2 \rho_1(\mathbf{r}_1) \frac{e^2}{4\pi\epsilon_0 r_{12}} \\
 & = \int_0^\infty dr_1 r_1^2 \rho_1(\mathbf{r}_1) V_{21}(\mathbf{r}_1) + \int_0^\infty dr_2 r_2^2 \rho_2(\mathbf{r}_2) V_{12}(\mathbf{r}_2).
 \end{aligned} \tag{B.2}$$

Figure B.1 illustrates the rearrangement of the ranges of integration. Here $V_{21}(\mathbf{r}_1)$ is the *partial* electrostatic potential at $\mathbf{r}_1$ caused by the charge density $\rho(\mathbf{r}_2)$, partial because it is caused by only that part of $\rho(\mathbf{r}_2)$ that lies at radii less than r_1 ; and a similar definition applies to $V_{12}(\mathbf{r}_2)$. The entire electrostatic energy J is now obtained by integrating expression B.2 over the angles θ_1 , ϕ_1 , θ_2 and ϕ_2 . We obtain

$$J = \int \rho_1(\mathbf{r}_1) V_{21}(\mathbf{r}_1) d^3\mathbf{r}_1 + \int \rho_2(\mathbf{r}_2) V_{12}(\mathbf{r}_2) d^3\mathbf{r}_2. \tag{B.3}$$

Expression B.3 can be applied to any charge densities, however complicated their dependence on the angles θ and ϕ . But the significance of our result is most easily explained if we take the special case where the potentials $V_{21}(\mathbf{r}_1)$ and $V_{12}(\mathbf{r}_2)$ are spherically symmetric—independent of angles—either in fact or as the result of imposing an approximation. In such a case we can think of $V_{21}(r_1)$ as the (partial) potential resulting from a radial electric field⁴ that is felt by electron 1 owing to the presence of electron 2; and likewise $V_{12}(r_2)$ is the (partial) potential felt

⁴In the trade, the potential itself is usually referred to as a radial, or *central* field. We should point out that the functions V_{12} and V_{21} are here not necessarily the same functions of their arguments; indeed, they will generally be different unless $\rho_1(r) = \rho_2(r)$. Later in this book we shall discuss *the* central field, which is a single $V(r)$, acting alike on all electrons. The potentials introduced here are different from this common-to-all central field.

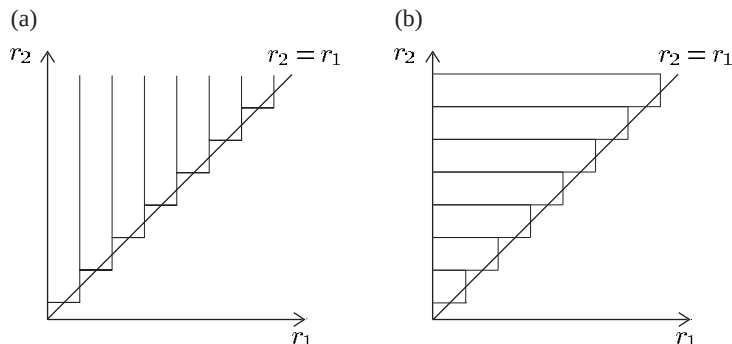


Fig. B.1 Integration over the region $r_2 > r_1 > 0$ can be carried out in two ways: (a) integration with respect to r_2 from $r_2 = r_1$ to ∞ , followed by integration from $r_1 = 0$ to ∞ ; or (b) integration with respect to r_1 from 0 to $r_1 = r_2$, followed by integration from $r_2 = 0$ to ∞ . The latter is convenient for calculating the electrostatic interaction in eqn B.2 and it leads to two contributions that are related by an interchange of the particle labels $r_1 \leftrightarrow r_2$. For a symmetrical configuration such as $1s^2$ in helium these two contributions are equal and this feature reduces the amount of calculation. By definition, exchange integrals are unchanged by swapping the labels $r_1 \leftrightarrow r_2$ and so the same comments apply to their evaluation.

by electron 2 owing to the presence of electron 1. When $\rho(\mathbf{r}_1) = \rho(\mathbf{r}_2)$, as in the calculation of the direct integral for the $1s^2$ configuration (Section 3.3.1), these two parts of eqn B.3 are equal (and only one integral needs to be calculated).⁵

⁵A similar simplification can be used in the calculation of exchange integrals, as in Exercise 3.6.

The fact that the potentials V_{12} and V_{21} are *partial* potentials deserves comment. Had we defined a $V_{21}(\mathbf{r}_1)$ to represent the whole of the electrostatic potential from $\rho_2(\mathbf{r}_2)$ acting on electron 1, then the interaction energy J would have been given by the single term

$$\begin{aligned} \int_0^\pi d\theta_1 \sin \theta_1 \int_0^{2\pi} d\phi_1 \int_0^\infty dr_1 r_1^2 \rho_1(\mathbf{r}_1) V_{21}(r_1) \\ = \int \rho_1(\mathbf{r}_1) V_{21}(\mathbf{r}_1) d^3\mathbf{r}_1, \end{aligned} \quad (\text{B.4})$$

or by a similar expression with the labels 1 and 2 interchanged throughout.⁶ Such a presentation is, of course, physically correct, but it is much less convenient mathematically than the version presented earlier; it precludes the separation of the Schrödinger equation into two equations, each describing the behaviour of one electron.

The simplified form for J given in eqn B.3 is used in the evaluation of direct and exchange integrals for helium in Chapter 3, and is generally applicable to electrostatic energies.

⁶The physical idea that each electron should ‘feel’ a potential due to the other is so intuitive that there can be a strong temptation to write the interaction energy down twice—once in the form B.4 and a second time with 1 and 2 interchanged. If this is done, the interaction energy is double-counted. The advantage of the form B.3 is precisely that it conforms to the intuitive idea without double-counting.

Appendix C: Magnetic dipole transitions



The electric dipole transitions that lead to the emission and absorption of light by atoms are discussed in many places in this book, and the selection rules for this type of transition are summarised in Table 5.1. These rules are not obeyed in radio-frequency transitions between Zeeman sub-levels where only the magnetic quantum number changes, or in transitions between different hyperfine levels where F and M_F can change—these transitions in radio-frequency spectroscopy are magnetic dipole or M1 transitions induced by the oscillating magnetic field of the radiation¹

$$\mathbf{B}_{\text{rf}} = \mathbf{B}_0 \cos \omega t. \quad (\text{C.1})$$

The transition matrix element between hyperfine levels is

$$\mu_{21} \propto \langle 2 | \boldsymbol{\mu} \cdot \mathbf{B}_{\text{rf}} | 1 \rangle, \quad (\text{C.2})$$

where $\boldsymbol{\mu}$ is the magnetic dipole operator in eqn 5.9. This gives transitions for which $\Delta l = \Delta L = \Delta S = 0$.

The selection rules for magnetic dipole transitions between hyperfine states are:

$$\begin{aligned} \Delta F &= 0, \pm 1 \quad (\text{but not } 0 \rightarrow 0), \\ \Delta M_F &= 0, \pm 1. \end{aligned}$$

These are as expected from angular momentum conservation and a dipole operator that can change angular momentum by one unit (as discussed for electric dipole transitions in Section 2.3.5).

The spontaneous decay rate for radio-frequency transitions is proportional to

$$A_{21} \propto \omega^3 |\mu_{21}|^2, \quad (\text{C.3})$$

where ω is the angular frequency, which is small relative to optical transitions. The matrix element μ_{12} is also much smaller than that for E1 transitions:

$$\frac{|\langle 2 | \boldsymbol{\mu} \cdot \mathbf{B} | 1 \rangle|^2}{|\langle 3 | e\mathbf{r} \cdot \mathbf{E} | 1 \rangle|^2} \sim \left(\frac{\mu_B/c}{ea_0/Z} \right)^2 \sim (Z\alpha)^2, \quad (\text{C.4})$$

where the factor c arises because the ratio of the magnetic to the electric field, $|\mathbf{B}|/|\mathbf{E}| = 1/c$ in an electromagnetic wave; the atomic size scales as $1/Z$, see Section 1.9. Therefore spontaneous emission is negligible in radio-frequency and microwave spectroscopy. In outer space, however,

¹An important exception is the Lamb shift transition, where two levels of opposite parity have a very small energy separation (see Section 2.3.4). This is an electric dipole or E1 transition.

the gas density is low and there are vast clouds of atomic hydrogen; even weak emission at a wavelength of 21 cm gives enough microwave radiation to be picked up by radio telescopes.

The selection rules for magnetic dipole radiation transitions are given in the table below, together with the electric dipole transitions for comparison.

Electric dipole transitions	Magnetic dipole transitions
$\Delta J = 0, \pm 1$ (but not $J = 0$ to $J' = 0$)	$\Delta J = 0, \pm 1$ (but not $J = 0$ to $J' = 0$)
$\Delta M_J = 0, \pm 1$	$\Delta M_J = 0, \pm 1$
Parity change	No parity change
$\Delta l = \pm 1$	$\Delta l = 0$ } No change of
Any Δn	$\Delta n = 0$ } configuration
$\Delta L = 0, \pm 1$	$\Delta L = 0$
$\Delta S = 0$	$\Delta S = 0$

In certain circumstances magnetic dipole transitions can give rise to visible transitions in atoms, and there are also electric quadrupole (E2) transitions arising from the breakdown of the dipole approximation. These forbidden transitions, i.e. forbidden for electric dipole transitions, are detailed in Corney (2000).

Appendix D: The line shape in saturated absorption spectroscopy

D

The description of saturated absorption spectroscopy in Section 8.3 explained qualitatively how this technique gives a Doppler-free signal. This appendix gives a more quantitative treatment based on modifying eqn 8.11 to account for the change in populations produced by the light. We shall use $N_1(v)$ to denote the number density of atoms in level 1 with velocities v to $v + dv$ (along the direction of light) and $N_2(v)$ to denote those in level 2 within the same velocity class. At low intensities most atoms remain in the ground state, so $N_1(v) \simeq Nf(v)$ and $N_2(v) \simeq 0$. Higher intensity radiation excites atoms close to the resonant velocity (given in eqn 8.12) into the upper level, as shown in Fig. 8.4. Within each narrow range of velocities v to $v + dv$ the radiation affects the atoms in the *same* way, so we can use eqn 7.82 for homogeneous broadening to write the difference in population densities (for atoms in a given velocity class) as

$$N_1(v) - N_2(v) = Nf(v) \times \frac{1}{1 + (I/I_{\text{sat}})L(\omega - \omega_0 + kv)}. \quad (\text{D.1})$$

This includes the Doppler shift $+kv$ for a laser beam propagating in the opposite direction to atoms with positive velocities, e.g. the pump beam in Fig. 8.4. The Lorentzian function $L(\omega - \omega_0 + kv)$ is defined so that $L(0) = 1$, namely

$$L(x) = \frac{\Gamma^2/4}{x^2 + \Gamma^2/4}. \quad (\text{D.2})$$

For low intensities $I \ll I_{\text{sat}}$, we can make the approximation

$$N_1(v) - N_2(v) \simeq Nf(v) \left\{ 1 - \frac{I}{I_{\text{sat}}} L(\omega - \omega_0 + kv) \right\}. \quad (\text{D.3})$$

The expression inside the curly brackets equals unity except near $v = -(\omega - \omega_0)/k$ and gives a mathematical representation of the ‘hole burnt’ in the Maxwellian velocity distribution $Nf(v)$ by the pump beam (as illustrated in Figs 8.3 and 8.4). In this low-intensity approximation the hole has a width of $\Delta v = \Gamma/k$. Atoms in each velocity class absorb light with a cross-section given by eqn 7.76, with a frequency detuning that takes into account the Doppler shift. The absorption of a weak probe

beam travelling along the z -direction through a gas with a strong pump beam in the opposite direction is given by the integral in eqn 8.17:

$$\begin{aligned}\kappa(\omega, I) &= \int \{N_1(v) - N_2(v)\} \sigma(\omega - kv) dv \\ &= \int \frac{Nf(v)}{1 + \frac{I}{I_{\text{sat}}} \frac{\Gamma^2/4}{(\omega - \omega_0 + kv)^2 + \Gamma^2/4}} \times \frac{\sigma_0 \Gamma^2/4}{(\omega - \omega_0 - kv)^2 + \Gamma^2/4} dv.\end{aligned}\quad (\text{D.4})$$

Note the opposite sign of the Doppler shift for the probe ($-kv$) and pump ($+kv$) beams. (Both beams have angular frequency ω in the laboratory frame.) For low intensities, $I/I_{\text{sat}} \ll 1$, the same approximation as in going from eqn D.1 to D.3 gives

$$\kappa(\omega, I) = N\sigma_0 \int f(v) L(\omega - \omega_0 - kv) \left\{ 1 - \frac{I}{I_{\text{sat}}} L(\omega - \omega_0 + kv) \right\} dv. \quad (\text{D.5})$$

As $I \rightarrow 0$, this reduces to eqn 8.11 for Doppler broadening without saturation, i.e. the convolution of $f(v)$ and $L(\omega - \omega_0 - kv)$. The intensity-dependent part contains the integral

$$\begin{aligned}\int_{-\infty}^{\infty} f(v) L(\omega - \omega_0 - kv) L(\omega - \omega_0 + kv) dv \\ = f(v=0) \int_{-\infty}^{\infty} \frac{\Gamma^2/4}{x^2 + \Gamma^2/4} \times \frac{\Gamma^2/4}{\{2(\omega - \omega_0) - x\}^2 + \Gamma^2/4} \frac{dx}{k}.\end{aligned}\quad (\text{D.6})$$

The product of the two Lorentzian functions is small, except where both $\omega - \omega_0 + kv = 0$ and $\omega - \omega_0 - kv = 0$. Solving these two equations we find that the integrand only has a significant value when $kv = 0$ and $\omega - \omega_0 = 0$. The Gaussian function does not vary significantly from $f(v=0)$ over this region so it has been taken outside the integral. The change of variables to $x = \omega - \omega_0 + kv$ shows clearly that the integral is the convolution of two Lorentzian functions (that represent the hole burnt in the population density and the line shape for absorption of the probe beam). The convolution of two Lorentzian functions of widths Γ and Γ' gives another of width $\Gamma + \Gamma'$ (Exercise 8.8). The convolution of two Lorentzian functions with the same width $\Gamma = \Gamma'$ in eqn D.6 gives a Lorentzian function of width $\Gamma + \Gamma' = 2\Gamma$ with the variable $2(\omega - \omega_0)$; this is proportional to $g_{\text{H}}(\omega)$, as defined in eqn 7.77 (see Exercise 8.8). Thus a pump beam of intensity I causes the probe beam to have an absorption coefficient of

$$\kappa(\omega) = N \times 3 \frac{\pi^2 c^2}{\omega_0^2} A_{21} g_{\text{D}}(\omega) \left\{ 1 - \frac{I}{I_{\text{sat}}} \frac{\pi \Gamma}{4} g_{\text{H}}(\omega) \right\}. \quad (\text{D.7})$$

The function in the curly brackets represents the reduced absorption at the centre of the Doppler-broadened line—it gives the peak in the probe beam intensity transmitted through the gas when $\omega = \omega_0$, as shown in

Fig. 8.4(b). This saturated-absorption signal comes from the velocity class of atoms with $v = 0$.

We assumed that $I \ll I_{\text{sat}}$ to obtain eqn D.3, but usually in experiments the pump beam has an intensity close to saturation to give a strong signal. The simple rate equation treatment becomes inaccurate around I_{sat} and a more sophisticated approach is required based on the optical Bloch equations, as described in Letokhov and Chebotaev (1977). Also, optical pumping out of the state that interacts with the radiation into other Zeeman sub-levels, or hyperfine levels, can be the dominant way of depleting the lower-level population $N_1(v)$. However, the line shape remains symmetrical about ω_0 , so saturation spectroscopy gives an accurate measurement of the atomic resonance frequency.

E

Appendix E: Raman and two-photon transitions

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¹More rigorous treatments can be found in quantum mechanics texts.

E.1 Raman transitions

This appendix gives an explanation of Raman (and two-photon transitions) by adapting the treatment of single-photon transitions given in Chapter 7—this approach gives much more physical insight than simply quoting the theoretical formulae from second-order time-dependent perturbation theory.¹ A Raman transition involves two laser beams with frequencies ω_{L1} and ω_{L2} , and the atom interacts with an electric field that has two frequency components:

$$\mathbf{E} = \mathbf{E}_{L1} \cos(\omega_{L1}t) + \mathbf{E}_{L2} \cos(\omega_{L2}t). \quad (\text{E.1})$$

A Raman transition between two atomic levels, labelled 1 and 2, involves a third atomic level, as shown in Fig. 9.20. This third level is labelled i for intermediate, but it is very important to appreciate that atoms are not really excited to level i . The treatment presented here emphasises that Raman transitions are fundamentally different from a process comprised of two single-photon transitions ($1 \rightarrow i$ followed by $i \rightarrow 2$). As in Section 9.8, we take the frequencies of the levels to be related by $\omega_i \gg \omega_2 > \omega_1$.

Firstly, we consider the perturbation produced by the light at ω_{L1} for the transition between levels 1 and i . This is the same situation as for a two-level atom interacting with an oscillating electric field that was described in Chapter 7, but here the upper level is labelled i instead of having the label 2 (and we write ω_{L1} rather than ω). For a weak perturbation, the lower state $|1\rangle$ has amplitude $c_1(0) = 1$ and the amplitude of $|i\rangle$, from eqn 7.14, is

$$c_i(t) = \frac{\Omega_{i1}}{2} \left[\frac{1 - \exp\{i(\omega_i - \omega_1 - \omega_{L1})t\}}{\omega_i - \omega_1 - \omega_{L1}} \right]. \quad (\text{E.2})$$

Here the Rabi frequency for the transition Ω_{i1} is defined in terms of $\mathbf{E}_{L1}$ as in eqn 7.12.² We define the difference between the laser frequency ω_{L1} and the frequency of the transition between levels 1 and i as

$$\Delta = \omega_1 + \omega_{L1} - \omega_i. \quad (\text{E.3})$$

From eqn 7.76 the wavefunction of the atom is

$$\Psi_n(\mathbf{r}, t) = e^{-i\omega_1 t} |1\rangle - \frac{\Omega_{i1}}{2\Delta} e^{-i\omega_i t} |i\rangle + \frac{\Omega_{i1}}{2\Delta} e^{-i(\omega_1 + \omega_{L1})t} |i\rangle. \quad (\text{E.4})$$

²We assume that the Rabi frequency is real, i.e. $\Omega_{i1}^* = \Omega_{i1}$.

The perturbation produces the admixture of two terms into the initial state $|1\rangle$ that both have the same small amplitude $\Omega_{i1}/|2\Delta| \ll 1$. We will find that the term with angular frequency ω_i represents real excitation to $|i\rangle$. The term with angular frequency $\omega_1 + \omega_{L1}$ corresponds to a virtual level, i.e. in the mathematics this term acts as if there is a level at an energy $\hbar\omega_{L1}$ above the ground state that has the symmetry properties of $|i\rangle$, but in reality there is no such level.

To determine the effect of the field oscillating at ω_{L2} on the perturbed atom, we take eqn 7.10, which states that for a single-photon transition $i\dot{c}_2 = \Omega \cos(\omega t) e^{i\omega_0 t} c_1$, and make the replacements $\omega \rightarrow \omega_{L2}$, $\omega_0 \rightarrow \omega_2 - \omega_i$ and $\Omega \rightarrow \Omega_{2i}$ to obtain³

$$i\dot{c}_2(t) = \Omega_{2i} \cos(\omega_{L2}t) e^{i(\omega_2 - \omega_i)t} c_i(t). \quad (\text{E.5})$$

Insertion of the expression for $c_i(t)$ from eqn E.2 yields

$$\begin{aligned} i\dot{c}_2(t) &= -\Omega_{2i} \cos(\omega_{L2}t) e^{-i(\omega_i - \omega_2)t} \times \frac{\Omega_{i1}}{2\Delta} \left[1 - e^{i(\omega_i - \omega_1 - \omega_{L1})t} \right] \\ &= -\frac{\Omega_{2i}\Omega_{i1}}{4\Delta} \left[e^{i\omega_{L2}t} + e^{-i\omega_{L2}t} \right] \cdot \left[e^{-i(\omega_i - \omega_2)t} - e^{i\{(\omega_2 - \omega_1) - \omega_{L1}\}t} \right]. \end{aligned} \quad (\text{E.6})$$

Integration and the rotating-wave approximation lead to

$$\begin{aligned} c_2(t) &= \frac{\Omega_{2i}\Omega_{i1}}{4\Delta} \left[\frac{1 - e^{-i(\omega_i - \omega_2 - \omega_{L2})t}}{\omega_i - \omega_2 - \omega_{L2}} + \frac{1 - e^{i\{(\omega_2 - \omega_1) - (\omega_{L1} - \omega_{L2})\}t}}{(\omega_2 - \omega_1) - (\omega_{L1} - \omega_{L2})} \right] \\ &= \frac{\Omega_{2i}\Omega_{i1}}{4\Delta(\Delta + \delta)} \left[1 - e^{-i(\Delta + \delta)t} \right] - \frac{\Omega_{2i}\Omega_{i1}}{4\Delta\delta} \left[1 - e^{-i\delta t} \right], \end{aligned} \quad (\text{E.7})$$

where

$$\delta = (\omega_{L1} - \omega_{L2}) - (\omega_2 - \omega_1) \quad (\text{E.8})$$

is the frequency detuning from the Raman resonance condition that the difference in the laser frequencies matches the energy difference between levels 1 and 2, over $\hbar$ (see Fig. 9.20).⁴ Equation E.7 looks complicated but its two parts have a straightforward physical interpretation—we can find the conditions for which each part is important by examining their denominators. Raman transitions are important when $\delta \simeq 0$ and Δ is large ($|\delta| \ll |\Delta|$) so that the second part of eqn E.7 dominates (and the individual single-photon transitions are far from resonance). Defining an effective Rabi frequency as

$$\Omega_{\text{eff}} = \frac{\Omega_{2i}\Omega_{i1}}{2\Delta} = \frac{\langle 2 | \mathbf{e} \cdot \mathbf{E}_{L2} | i \rangle \langle i | \mathbf{e} \cdot \mathbf{E}_{L1} | 1 \rangle}{\hbar^2 (\omega_i - \omega_1 - \omega_{L1})}, \quad (\text{E.9})$$

we can write eqn E.7 as

$$c_2(t) = \frac{\Omega_{\text{eff}}}{2} \frac{1 - e^{-i(\Delta + \delta)t}}{\Delta + \delta} - \frac{\Omega_{\text{eff}}}{2} \frac{1 - e^{-i\delta t}}{\delta}. \quad (\text{E.10})$$

The first term can be neglected when $|\delta| \ll |\Delta|$ to yield

$$|c_2(t)|^2 = \frac{1}{4} \Omega_{\text{eff}}^2 t^2 \text{sinc}^2\left(\frac{\delta t}{2}\right). \quad (\text{E.11})$$

³The possibility that radiation at angular frequency ω_{L1} could drive this transition between i and 2 is considered later for two-photon transitions.

⁴Notice that this condition does not depend on ω_i . The Raman transition can be viewed as a coupling between $|1\rangle$ and $|2\rangle$ via a virtual level, whose origin can be traced back to the term (with a small amplitude) at frequency $\omega_1 + \omega_{L1}$ in eqn E.4. However, although a virtual level gives a useful physical picture it is entirely fictitious—during a Raman transition there is negligible population in the excited state and hence negligible spontaneous emission.

This is the same as eqn 7.15 for one-photon transitions, but Ω_{eff} has replaced Ω . This presentation of Raman transitions has assumed a weak perturbation and we have found results analogous to those for the weak excitation of a single-photon transition (Section 7.1); a more comprehensive treatment of the Raman coupling between $|1\rangle$ and $|2\rangle$ with effective Rabi frequency Ω_{eff} , analogous to that in Section 7.3, shows that Raman transitions give rise to Rabi oscillations, e.g. a π -pulse transfers all the population from 1 to 2, or the reverse. Raman transitions are coherent in the same way as radio-frequency, or microwave, transitions directly between the two-levels, e.g. a Raman pulse can put the atomic wavefunction into a coherent superposition state $A|1\rangle + B|2\rangle$.⁵

⁵Raman transitions can impart momentum to the atoms and this makes them extremely useful for manipulating atoms and ions (see Section 9.8).

It is vital to realise that the Raman transition has a quite distinct nature from a transition in two steps, i.e. a single-photon transition from level 1 to i and then a second step from i to 2. The two-step process would be described by rate equations and have spontaneous emission from the real intermediate state. This process is more important than the coherent Raman process when the frequency detuning Δ is small so that ω_{L1} matches the frequency of the transition between $|1\rangle$ and $|i\rangle$.⁶ The distinction between a coherent Raman process (involving simultaneous absorption and stimulated emission) and two single-photon transitions can be seen in the following example.

Example E.1 The duration of a π -pulse (that contains both frequencies ω_{L1} and ω_{L2}) is given by

$$\Omega_{\text{eff}} t_{\pi} = \pi. \quad (\text{E.12})$$

For simplicity, we shall assume that both Raman beams have similar intensities so that $\Omega_{i1} \simeq \Omega_{2i} \simeq \Omega$ and hence $\Omega_{\text{eff}} \simeq \Omega^2/\Delta$ (neglecting small factors).⁷ From eqn 9.3 we find that the rate of scattering of photons on the transition $|1\rangle$ to $|i\rangle$ is approximately $\Gamma\Omega^2/\Delta^2$, since for a Raman transition Δ is large ($\Delta \gg \Gamma$). Thus the number of spontaneously-emitted photons during the Raman pulse is

$$R_{\text{scatt}} t_{\pi} \simeq \frac{\Gamma\Omega^2}{\Delta^2} \frac{\pi\Delta}{\Omega^2} \simeq \frac{\pi\Gamma}{\Delta}. \quad (\text{E.13})$$

This shows that spontaneous emission is negligible when $\Delta \gg \Gamma$. As a specific example, consider the Raman transition between the two hyperfine levels in the 3s ground configuration of sodium⁸ driven by two laser beams with frequencies such that $\Delta = 2\pi \times 3 \text{ GHz}$. The $3p \ ^2P_{1/2}$ level acts as the intermediate level and has $\Gamma = 2\pi \times 10^7 \text{ s}^{-1}$. Thus $R_{\text{scatt}} t_{\pi} \simeq 0.01$ —this means that the atoms can be subjected to many π -pulses before there is a spontaneous emission event that destroys the coherence (e.g. in an interferometer as described in Section 11.7). Admittedly, this calculation is crude but it does indicate the relative importance of the coherent Raman transitions and excitation of the intermediate level by single-photon processes. (It exemplifies the order-of-magnitude estimate that should precede calculations.) The same ap-

⁶Similarly, the quantity $\Delta + \delta = \omega_i - \omega_2 - \omega_{L2}$ (in eqn E.7) is small when ω_{L2} matches the frequency of the transition between $|i\rangle$ and $|2\rangle$. For this condition the single-photon transition between levels i and 2 is the dominant process. The single-photon processes can be traced back to the small amplitude with time dependence $\exp(i\omega_i t)$ in eqn E.4.

⁷The generalisation to the case where $\Omega_{i1} \neq \Omega_{2i}$ is straightforward.

⁸The hyperfine splitting is $\omega_2 - \omega_1 = 2\pi \times 1.7 \text{ GHz}$ but we do not need to know this for this calculation.

proximations give the duration of a π -pulse as

$$t_\pi = \frac{\pi}{\Omega_{\text{eff}}} \simeq \frac{\pi\Delta}{\Omega^2} \simeq \frac{2\pi\Delta}{\Gamma^2} \frac{I_{\text{sat}}}{I}, \quad (\text{E.14})$$

where eqn 7.86 has been used to relate Ω^2 to intensity. Thus in a Raman experiment carried out with two laser beams, each of intensity $3I_{\text{sat}}$, and the above values of Δ and Γ for sodium, the pulse has a duration of $t_\pi = 10 \mu\text{s}$.

E.2 Two-photon transitions

The intuitive description of Raman transitions as two successive applications of the first-order time-dependent perturbation theory result for a single-photon transition can also be applied to two-photon transitions between levels 1 and 2 via i , where $\omega_2 > \omega_i > \omega_1$. The two-photon rate between levels 1 and 2 is

$$R_{12} = \left| \sum_i \left\{ \frac{\langle 2 | e\mathbf{r} \cdot \mathbf{E}_{L2} | i \rangle \langle i | e\mathbf{r} \cdot \mathbf{E}_{L1} | 1 \rangle}{\hbar^2 (\omega_i - \omega_1 - \omega_{L1})} + \frac{\langle 2 | e\mathbf{r} \cdot \mathbf{E}_{L1} | i \rangle \langle i | e\mathbf{r} \cdot \mathbf{E}_{L2} | 1 \rangle}{\hbar^2 (\omega_i - \omega_1 - \omega_{L2})} \right\} \right|^2 \times g(\omega_{L1} + \omega_{L2}). \quad (\text{E.15})$$

This has the form of the modulus squared of a sum of amplitudes multiplied by the line shape function $g(\omega_{L1} + \omega_{L2})$. There are two contributing amplitudes from (a) the process where the atom interacts with the beam whose electric field is $\mathbf{E}_{L1}$ and then with the beam whose field is $\mathbf{E}_{L2}$, and (b) the process where the atom absorbs photons from the two laser beams in the opposite order.⁹ The energy increases by $\hbar(\omega_{L1} + \omega_{L2})$ independent of the order in which the atom absorbs the photons, and the amplitude in the excited state is the sum of the amplitudes for these two possibilities. In Doppler-free two-photon spectroscopy (Section 8.4) the two counter-propagating laser beams have the same frequency $\omega_{L1} = \omega_{L2} = \omega$, and we shall also assume that they have the same magnitude of electric field (as would be the case for the apparatus shown in Fig. 8.8). This leads to an excitation rate given by

$$R_{12} \simeq \left| 2 \sum_i \frac{\Omega_{2i}\Omega_{i1}}{\omega_i - \omega_1 - \omega} \right|^2 \cdot \frac{\tilde{\Gamma}/(2\pi)}{(\omega_{12} - 2\omega)^2 + \tilde{\Gamma}^2/4}, \quad (\text{E.16})$$

with Ω_{i1} and Ω_{2i} as defined in the previous section. The transition has a homogeneous width $\tilde{\Gamma}$ greater than, or equal to, the natural width of the upper level $\tilde{\Gamma} \geq \Gamma$; this Lorentzian line shape function has a similar form to that in eqn 7.77, with a maximum at the two-photon resonance frequency $\omega_{12} = \omega_2 - \omega_1$ (as in Section 8.4). The constraint that the two photons have the same frequency means that the frequency detuning from the intermediate level $\Delta = \omega_i - (\omega_1 + \omega)$ is generally much larger than in Raman transitions, e.g. for the 1s–2s transition in

⁹Only one of these paths is near resonance for Raman transitions because $\omega_{L1} - \omega_{L2} \neq \omega_{L2} - \omega_{L1}$.

hydrogen the nearest level than can provide an intermediate state is 2p, which is almost degenerate with the 2s level (see Fig. 8.4); thus $\omega_i \simeq \omega_2$ and $\Delta \simeq \omega = 2\pi \times 10^{15} \text{ s}^{-1}$ (or 4×10^5 times larger than the frequency detuning used in the example of a Raman transition in the previous section). There are two important consequences of this large frequency detuning: (a) in real atoms there are many levels with comparable frequency detuning and taking these other paths ($1 \rightarrow i \rightarrow 2$) into account leads to the summation over i in eqn E.16; and (b) the rate of two-photon transitions is small even for high intensities (cf. allowed single-photon transitions).¹⁰

¹⁰A rough estimate of the two-photon rate can be made in a similar way to the calculations for Raman transitions in the previous section (Dentröder 1996).

Appendix F: The statistical mechanics of Bose–Einstein condensation



This appendix does not attempt to reproduce the standard treatment of a Bose–Einstein condensate found in statistical mechanics texts, but aims to give a complementary viewpoint, that emphasises the link between photons and atoms, and also to describe BEC in a harmonic potential.

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F.1 The statistical mechanics of photons

The Planck formula for the energy density of radiation per unit bandwidth $\rho(\omega)$, used in the Einstein treatment of radiation (see eqn 1.29), can be written as a product of three factors:

$$\rho(\omega) d\omega = \hbar\omega \times f_{\text{ph}}(\omega) \times D_{\text{ph}}(\omega) d\omega. \quad (\text{F.1})$$

Here $\hbar\omega$ is the photon energy, the function $f_{\text{ph}}(\omega) = 1/(e^{\beta\hbar\omega} - 1)$, with $\beta = 1/k_{\text{B}}T$, determines the number of photons per energy level and $D_{\text{ph}}(\omega)$ is the density of states per unit bandwidth.¹ Although the distribution f_{ph} becomes very large as $\omega \rightarrow 0$ (infra-red divergence), the integration over the frequency distribution (using the substitution $x = \beta\hbar\omega$) yields a finite result for the total energy of the radiation in the volume V :

$$E = V \int_0^\infty \rho(\omega) d\omega \propto VT^4. \quad (\text{F.2})$$

This result follows from dimensional considerations, without the evaluation of the definite integral.² This integral for E is a particular case of the general expression in statistical mechanics for the energy of the system which is obtained by summing the energy over all occupied levels:

$$E = \sum_i f(\varepsilon_i) \varepsilon_i. \quad (\text{F.3})$$

Here $f(\varepsilon_i)$ gives the distribution over the levels of energy ε_i . The integral in eqn F.2, for the particular case of photons, gives a close approximation

¹The number of states in phase-space with wavevectors between k and $k + dk$ equals the volume of a spherical shell of thickness dk times the density of the states in k -space, $4\pi k^2 dk \times V/(2\pi)^3$. For photons we need an extra factor of 2, because of the different possible polarizations, and the substitution $k = \omega/c$.

²The energy density E/V has the same T^4 dependence as the Stefan–Boltzmann law for the power per unit area radiated by a black body (as expected, since cE/V corresponds to power divided by area).

to this summation when the system occupies many levels to give an almost continuous distribution. The density of states in the integral represents a summation over the number of levels within each frequency (or energy) interval $d\omega$ ($\equiv d\varepsilon/\hbar$).

F.2 Bose–Einstein condensation

The crucial difference between photons and particles in statistical mechanics is that photons simply disappear as the temperature tends to zero, as we can see from eqn F.2. In contrast, for a system of particles, such as a gas of atoms or molecules in a box, the number remains constant. This introduces a second constraint on the distribution function, namely that the sum of the populations in all the energy levels equals the total number $\mathcal{N}$. Note that the symbol ‘ $\mathcal{N}$ ’ is used here because ‘ N ’ was used in Chapter 7 to represent the number density (the convention usually adopted in laser physics). Here, number density is denoted by ‘ n ’, as is usual in statistical mechanics. Hence

$$\mathcal{N} = \sum_i f(\varepsilon_i). \quad (\text{F.4})$$

This equation for conservation of number, and eqn F.3 for the total energy, apply to any system of particles. We shall consider particles with integer spin that follow the Bose–Einstein distribution function

$$f_{\text{BE}}(\varepsilon) = \frac{1}{e^{\beta(\varepsilon-\mu)} - 1}. \quad (\text{F.5})$$

This function has two parameters β and μ that can be determined by the two constraints. For the particular case of bosons at low temperatures, the chemical potential μ has little consequence *except* for atoms in the lowest energy state, so that for the higher-lying levels $f_{\text{BE}}(\varepsilon)$ closely resembles the distribution of photons! We can justify this assertion by considering the properties of a system with a significant population $\mathcal{N}_0$ in the ground state.³ The number of atoms in the lowest level with energy ε_0 is given by

$$\mathcal{N}_0 = \frac{1}{e^{\beta(\varepsilon_0-\mu)} - 1}. \quad (\text{F.6})$$

Hence

$$\frac{\varepsilon_0 - \mu}{k_{\text{B}}T} = \ln \left(1 + \frac{1}{\mathcal{N}_0} \right) \simeq \frac{1}{\mathcal{N}_0}. \quad (\text{F.7})$$

Einstein originally considered gases with a large total number of atoms, $\mathcal{N} \sim 10^{23}$, so, even when $\mathcal{N}_0$ is only a small fraction of $\mathcal{N}$, the difference $\varepsilon_0 - \mu$ is negligible in comparison to the thermal energy $k_{\text{B}}T$. (This thermodynamic, or large number, approximation works even for samples of 10^6 magnetically-trapped atoms.⁴) The equation shows that the chemical potential is lower than ε_0 , the lowest energy level in the system.⁵ For the first excited level we find that

$$\varepsilon_1 - \mu = (\varepsilon_1 - \varepsilon_0) + (\varepsilon_0 - \mu) \simeq \hbar\omega + \frac{k_{\text{B}}T}{\mathcal{N}_0} \simeq \hbar\omega.$$

³This may appear to be a circular argument since the large occupation of the lowest level is the signature of the Bose–Einstein condensation that we want to investigate! This section shows that this is a consistent solution of the equations for bosons, and the validity of this treatment can be appreciated more readily after deriving the equations.

⁴We assume a thermal energy $k_{\text{B}}T$ much greater than the spacing between the energy levels. Otherwise, particles sit in the ground state simply because of the Boltzmann factor $\exp\{-\beta(\varepsilon_1 - \varepsilon_0)\} \ll 1$, where ε_1 is the energy of the first excited level (Section 12.9).

⁵Otherwise there would be an energy for which $\varepsilon - \mu = 0$; this would make the denominator in eqn F.5 equal to zero so that $f(\varepsilon) \rightarrow \infty$.

Hence, *except for the ground-state population*, μ can be neglected and f_{BE} becomes the same as the distribution for photons:

$$f(\varepsilon) \simeq \frac{1}{e^{\beta\varepsilon} - 1}. \quad (\text{F.8})$$

From what has been said above you may wonder how neglecting the chemical potential can be consistent with conservation of particle number in eqn F.4. This equation can be expressed as an integral of $f(\varepsilon)$ times the density of states *for particles*, $D(\varepsilon)$,

$$\mathcal{N} = \mathcal{N}_0 + \int_0^\infty f(\varepsilon) D(\varepsilon) d\varepsilon. \quad (\text{F.9})$$

The number in the ground state, $\mathcal{N}_0$, has to be put in explicitly because the integral does not properly count these atoms. Effectively, we have replaced μ as a parameter by $\mathcal{N}_0$ (these are related by eqn F.7). The two terms in eqn F.9 give the number of particles in the two parts, or sub-systems, that make up the whole. From this perspective we regard the $\mathcal{N} - \mathcal{N}_0$ particles in the excited states ($\varepsilon > \varepsilon_0$) as a sub-system that exchanges particles with the condensate (atoms in the ground state). Thus atoms in the excited states behave as if there is no number conservation: $\mathcal{N} - \mathcal{N}_0 \rightarrow 0$ when $T \rightarrow 0$, as for photons.

The integral in eqn F.9 contains the distribution function from eqn F.8 times the density of states for particles given by

$$D(\varepsilon) = AV\varepsilon^{1/2} d\varepsilon, \quad (\text{F.10})$$

where A is a constant.⁶ With the substitution $x = \beta\varepsilon$, eqn F.9 becomes

$$\mathcal{N}_0 = \mathcal{N} - AV(k_{\text{B}}T)^{3/2} \zeta, \quad (\text{F.11})$$

where ζ represents the value of the integral given in statistical mechanics texts as

$$\zeta = \int_0^\infty \frac{x^{1/2}}{e^x - 1} dx = 2.6 \times \frac{\sqrt{\pi}}{2}. \quad (\text{F.12})$$

The ground-state occupation goes to zero, $\mathcal{N}_0 = 0$, at the critical temperature T_{C} given by

$$\frac{\mathcal{N}}{V} = A(k_{\text{B}}T_{\text{C}})^{3/2} \zeta. \quad (\text{F.13})$$

With $A = 2\pi(2M)^{3/2}/h^3$ and eqn F.12 for ζ , this gives eqn 10.14. The discussion here supposes that there is a large population in the lowest level (the Bose–Einstein condensate) and determines the temperature at which $\mathcal{N}_0$ goes to zero. (A different perspective adopted in many treatments is to consider what happens as atoms are cooled down towards T_{C} .) Dividing eqn F.11 by F.13 gives the fraction of particles in the ground state for a Bose gas in a box as

$$\frac{\mathcal{N}_0}{\mathcal{N}} = 1 - \left(\frac{T}{T_{\text{C}}} \right)^{3/2}. \quad (\text{F.14})$$

⁶ $D(\omega)$ differs fundamentally from $D_{\text{ph}}(\omega)$ in eqn F.1 because a particle's energy is proportional to the square of its wavevector, $\varepsilon \propto k^2$, i.e. $\varepsilon = p^2/2M$ with momentum $p = \hbar k$.

Note that the strength of the interaction between the atoms does not appear in this treatment—the value of T_C does not depend on the scattering length. This shows that *BEC arises from quantum statistics*. In real experiments there must be interactions so that atoms have a finite collision cross-section, otherwise there would not be any mechanism for establishing thermal equilibrium and evaporative cooling would not be possible. (A non-interacting Bose gas has some curious properties.)

F.2.1 Bose–Einstein condensation in a harmonic trap

The volume of the trapped atomic cloud depends on temperature as $V \propto T^{3/2}$ (from eqn 10.16); hence we find that for a trapped atom the equation equivalent to eqn F.11 is

$$\mathcal{N} - \mathcal{N}_0 \propto T^3. \quad (\text{F.15})$$

This dependence on the cube of T arises because the density of states for particles in a harmonic trap is different to that given in eqn F.10 for a gas in a box of fixed volume (i.e. an infinite square-well potential). This affects the way that the states fill up and hence the conditions for BEC. An argument analogous to that leading to eqn F.14 gives the fraction in the ground state as

$$\frac{\mathcal{N}_0}{\mathcal{N}} = 1 - \left(\frac{T}{T_C} \right)^3. \quad (\text{F.16})$$

This is a stronger dependence on T/T_C than in a homogeneous gas. At $T = 0.99 T_C$ this equation predicts a condensate fraction of $\mathcal{N}_0/\mathcal{N} = 0.03$, so that even just below T_C a cloud of $\mathcal{N} \sim 10^6$ trapped atoms gives $1/\mathcal{N}_0 \ll 1$, and this partly justifies the assumptions made after eqn F.7. Typically, experiments are carried out at around $T/T_C \sim 0.5$, or below, where only a fraction $(0.5)^3 = 0.125$ of the atoms remain in the thermal cloud. This gives a sufficiently pure condensate for most purposes and further evaporative cooling would cut deeply into the condensate and reduce $\mathcal{N}_0$.⁷

⁷A large condensate has a chemical potential that is considerably greater than the energy of the harmonic oscillator ground state; however, this turns out not to seriously affect results such as eqn F.16.

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